
XplainTM: Explainable Cost Sensitive Machine Learning for Anti Money Laundering Monitoring

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This section acknowledges the AI tools that were utilized in the process of completing this assignment.

Tool Name	Brief Description	Link to tool
N/A	N/A	N/A

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Abstract

Anti money laundering (AML) and Know Your Customer (KYC) compliance are 2 of the most expensive tasks for financial institutions. However, a large portion of the expense results from all the false positives that rule based monitoring systems produce rather than from an inability to identify suspicious activities. The real suspicious cases are sometimes overlooked because analysts spend most of their working week examining signals that prove to be entirely legitimate. This research represents study of XplainTM which is a machine learning pipeline that uses cost sensitive XGBoost classification and explanations from SHAP that reduce false positives while remaining understandable for regulators and compliance officers. We evaluate the pipeline using the IBM Transactions for Anti Money Laundering data set (Altman et al., 2023) and contrast it with a simple logistic regression model and a rules engine constructed using typology. The goal is not only to increase the number of suspicious cases but to create a system that works in real life and informs investigators of the reasons behind the cases escalation rather than just about the dependency on score.

Keywords: AML; XAI; Imbalanced classification; False positive reduction; RegTech; SHAP

1. Introduction

Beyond mere adherence Global banking systems are thought to be used to launder 2 to 5 percent of the world's GDP annually or between USD 800 billion and USD 2 trillion (Financial Action Task Force, 2022). In addition to harming financial organizations reputations money laundering can result in fines, penalties, suspension and even jail period for management. The amount of money being spent to achieve regulatory compliance is a result of the tremendous demand to act morally in terms of Anti Money Laundering (AML). Over USD 274 billion was spent on AML worldwide in 2022 (LexisNexis Risk Solutions, 2022).

The major issue is that large sums of money do not lead to the identification of money launderers. Every week it requires analysts to sort through thousands of false warnings. All too often, rule based transaction monitoring systems that issue warnings based on preprogrammed pattern search criteria or fixed thresholds are the problematic systems. Oztas et al. (2024) found that the systems false positive rate is usually more than 90% after interviewing 8 AML industry practitioners. The true alarms that the analysts are looking at are frequently on pages 2 or 3 of the wheel barrow of cases on their desks which is a big problem.

As a result, both the academic and professional communities are moving toward machine learning. Instead of following the guidelines that a committee or individual wrote 10 years ago lets use the transaction data to train a machine learning model that uses Artificial Intelligence technology to identify patterns in the data and more crucially where they don't exist. Gradient boosted tree models such as XGBoost model (Chen and Guestrin, 2016) have proven effective in classifying imbalanced tabular data. Regulators are interested in learning how and why transactions were found and also broker accountants have to defend their decisions. In other words, regardless of how successful other detection methods are defined as a model that merely assigns scores to every transaction but is unable to explain them is essentially worthless to regulators.

In comparison to a standard typology based rules baseline model on the IBM Transactions for Anti Money Laundering HI Small dataset *"Can a cost sensitive and explainable machine learning approach improve the task of AML transaction monitoring"* in terms of increasing true positives of fraudulent transactions and lowering false positives at a constant rate of alerts?

Is it possible to transfer model explanations utilizing SHAP (Lundberg and Lee, 2017) into regulatory acceptable reason codes that may be utilized to satisfy audit and investigation requirements without compromising the models performance?

Using the IBM Transactions for Anti Money Laundering HI Small dataset (Altman et al., 2023) this study will compare a cost sensitive XGBoost based model with a typology based rules system and a logistic regression model. This project will investigate if cost

sensitive learning improves performance and reduces false positive alarms at a predetermined number of triggered alerts as well as whether a SHAP explanation may provide practical and understandable regulator compliant reason codes that can be utilized for audit and investigation. Additionally, it will evaluate the effectiveness of these reason codes in providing analysts with the justifications for a recall. The goal is to determine whether explainable machine learning can perform better and more effectively than rules based methods for AML monitoring.

The rest of the suggestions follows through the shortcomings of AML systems machine learning for financial fraud detection, biasness, explainability in finance and regulatory technology (regtech) sector are all covered in Section 2 critical evaluation with ten important research publications. More information on the project approach, data, pipeline stages, tools to be utilized, evaluation, ethics and schedule is given in Section 3.

2. Literature Review

2.1. Rules based AML monitoring and its limitations

Let's examine what isn't working in order to comprehend what this research seeks to fix. The industry has been using rules based transaction monitoring for decades and it works well for a number of reasons including ease of explanation, ease of auditing and ease of comprehension for compliance groups at every stage. Unfortunately, the industry finds their cost to be increasingly intolerable.

One of the better recent analyses of the issue from the viewpoint of AML practitioners is the work published by [Oztas et al. \(2024\)](#) as they used semi structured interviews to conduct eight interviews and they found a common complaint as they don't have too many false positives, they understand that's an unfortunate but necessary aspect of the work. There will be less time to look into the intriguing plus highly suspicious items if the analysts spend the majority of their time on cases that are most likely uninteresting from the standpoint of money laundering. The interviewees were well aware that using machine learning would probably help with the false positive issue but they also firmly believed that any application of machine learning would need to be an auditable, explainable and helpful rather than merely faster and better.

[Alarab, Prakoonwit and Nacer's \(2022\)](#) discussion of using machine learning to detect money laundering in bitcoin blockchain transactions offers an intriguing perspective on the issue. They demonstrate that rules are inaccurate due to their inability to capture complex typologies as well as the large number of false positives. Rules are unable to record network activity and AML is typically multihop because money moves from one account to multiple others before being laundered back into the system. Using a graph convolutional network that identified features not covered by the rules, [Alarab et al. \(2022\)](#) initially showed that their data had an AUC of 0.97. Although their work focuses on blockchains rather than banks as the principle is the same money laundering exploits

the fact that regulations can only identify what is coded in. Together, Oztas et al. (2024) and Alarab et al. (2022) laid the groundwork for several machine learning based strategies which are discussed in the next two subsections.

2.2. Machine learning for AML detection

After making the case for alternatives to rules we must think about the kind of machine learning that could be used. Gradient boosted decision trees (GBTs) were the most accurate for classification density utilizing tabular data until the XGBoost method was introduced. Chen and Guestrin (2016) published a paper of the class imbalance “*scale-posweight*” can be taken into account to treat false negatives as more expensive we missed laundering than false positives we caught laundering which is crucial for the application of AML detection even though XGBoost is typically the most accurate. This study uses XGBoost as the primary classifier since AML requires this degree of versatility.

A more detailed theory is put out by Jensen and Iosifidis (2023). They suggest keeping the rules framework in place while using a machine learning triage model to assist prioritize the outcomes. The triage algorithm lifted 75 more high priority alerts from the rules and eliminated 33% of the false positives in the data they acquired from the bank. What a wonderful result and evidence of the usefulness of applying ML to supplement the rules system in the actual world. But their document lacks on the “*Explainability*”. It is difficult to assess whether their method would be helpful in a regulated setting where explanations are required since they score without offering justifications. Here’s where XplainTM can help it as it can provide the necessary explainability to make it feasible as well as the performance improvements reported by Jensen and Iosifidis (2023).

2.3. Imbalanced learning and cost sensitive methods

The issue of unbalanced learning may be more prevalent among people involved in financial crime. Approximately 0.12% of transactions in the IBM anti money laundering dataset used in this study are suspect. The majority of machine learning classifiers which strive for the most accurate categorization which accomplish this in this manner as it is quite simple to obtain 99.88% accuracy by assuming that everything is clean but they might not be particularly helpful for finding intriguing objects. According to Chawla et al. (2002) SMOTE improves the amount of positives in the data set by generating interpolating minority class examples from other minority class examples.

The idea is solid and Chawla et al. (2002) provided strong evidence that it is superior to straightforward random over sampling. The problem is that it can be noisy to resample and interpolate between minority class examples to create more minority class examples, especially if the minority class and majority class overlap as they do in AML where the most crucial requirements for laundering illicit money are that it should be blended with legitimate transactions.

Bahnsen et al. (2016) suggest that including words in the objective function that represent costs for various error costs is a better way to reflect the class imbalance in the sample

than altering the data to skew the class distributions in the training data. In terms of regulations, reputation or even money the cost of overlooking a transaction that is a part of a money laundering operation is higher than the cost of misidentifying one as cost sensitive learning adopts this realization. Specifically, they demonstrate that cost sensitive approaches perform better than resampling methods when the costs are known and constant. The comparison between Chawla et al. (2002) and Bahnsen et al. (2016) is not merely theoretical for this study because it supports one of the main research questions. Does SMOTE outperform cost sensitive XGBoost training in producing more operationally useful results in the IBM dataset? This can be considered a contribution of this research since it hasn't been explored before for the AML data.

2.4. Explainable AI and reasonable codes

Half the fight is creating a model that can reliably identify money laundering. Explaining the models predictions in order to act upon them is the other and potentially more challenging in practice. Adadi and Berrada (2018) provide a well summary of explainable AI models they separate the approaches into two categories, explainable post hoc approaches which are used after the black box model has been trained and inherently interpretable approaches. For the purposes of this discussion, it is their understanding that this is more than just a technical issue. The elephant in the room in XAI research to date is what is good enough for one user as a data scientist, compliance analyst and regulator will have different needs because it has been thought that the post hoc technical approaches can be left to handle the heavy work.

SHAP (Lundberg and Lee, 2017) is the particular type of XAI used here theoretically as SHAP has values that apply cooperative game theory and meet specific consistency and additivity criteria which makes it more appealing than alternatives like LIME. This implies that explanations provided for the same result in comparable circumstances will be more consistent overall which is a good quality if you're attempting to audit explanations. An exact as opposed to approximation Shapley value can be computed in a tractable manner using the TreeSHAP value of the SHAP model which is used here with XGBoost.

Together, Adadi and Berrada (2018) and Lundberg and Lee (2017) fail to provide compliance with the necessary layer which is transforming the model explanations that SHAP provides feature importance measures into narrative form as the reason code to be inserted in a SAR. This is the step when it will convert a model explanation into the narrative needed in an audit report. This is where the special XplainTM contribution comes in and current research that recognizes this step in an AML context does not offer it.

2.5. RegTech and auditability in financial crime monitoring

Arthur Arner, Elena Barberis and Ross Buckley (2017) are the ones who should map the evolution of RegTech to three waves of financial regulation since it is crucial that any research of this kind be conducted inside the framework of RegTech. They contend in a paper published that we are in a transitional stage, both in terms of how financial and other entities use technology for self regulation and in terms of how technology is used

in regulatory practice. This researchs four mainstreaming criteria for machine learning accuracy, auditability, consistency and regulatory fit might be viewed as part of the query's prescription. These four were taken into consideration when creating XplainTM.

The IBM Transactions for Anti Money Laundering (HI-Small) dataset which was made available by Altman et al. (2023) serves as the empirical basis for the study. The absence of labelled data for e.g., banks not providing their data and the few synthetic datasets available in AML research were either unrealistic or lacked typology labels making it difficult to perform quality AML research up until this point. Until Altman et al. (2023) enable accurate training and testing on data from the past and into the future, respectively, this dataset contains 550,000 synthetic transactions, 38 transaction typologies and timestamps. It has a positive class prevalence of 0.12%, which is comparable to what is found in the real world. This fact along with the data is publicly accessible and has been thoroughly examined in terms of typologies makes it a perfect fit for the work presented in this paper. Additionally, the fact that the data is accompanied by a paper that explains how the data was generated allows the work presented here to be contextualized and easily compared to other works.

2.6. Research gap and expected contribution

All things considered, the literature indicates that three things have not yet been tried. Firstly, a lot of research on machine learning for AML performance metrics such as AUC-ROC has been published but very little of it has examined performance at the set capacity level for alerts that banks and other financial institutions have. Secondly, although SHAP reasons for other financial crimes have been explored as there hasn't been any empirical research on combining explanations to identify standardized reason codes for AML. Thirdly, no comparison of cost sensitive XGBoost training and SMOTE resampling of a publicly available AML test data set with typology error analysis has been published. The three voids in an integrated end-to-end system are filled by XplainTM.

3. Methodology and Specifications

3.1. System Description

First and foremost, XplainTM is a transaction detection system that takes in raw transaction data and generates a list of risks that have been scaled and classified. Each risk is also accompanied by simply comprehensible reason codes that explain why the risk occurred. The procedure is a five step pipeline that leads to the next step which are load and split the data with no leakages, engineering behavioural features relevant to AML, building baseline comparators, training and optimizing the XGBoost pipeline model with cost sensitive learning and explain plus collect explanation based reason codes from SHAP. All of this is done on a laptop using free open source tools and in line Python in Jupyter Notebooks by importing state of the art compliance tools as the outcome is a CSV report that resembles a compliance decision making report.

3.2. The Dataset

The IBM Transactions for Anti Money Laundering HI Small dataset (Altman et al., 2023) served as the foundation for all of the empirical findings. This dataset contains about 500,000 financial transactions that have been classified as money laundering transactions and are associated with one or more of the 38 money laundering patterns. About 0.12% of the data contains money laundering which is what actually occurs in financial institutions. Since all of the data is dated we may use a temporal date split for training, validation and testing by 70%, 15% and 15% instead of a random train, valid and test split, which may cause problems with temporal data leakage. Any features that are aggregations over the previous N days will be calculated by looking before the date and time of a bank transaction because doing so is a somethings not leaking kind of calculation that will result in exaggerated detection rates but a useless detector.

3.3. Methodical Approach

Since this is a creation exercise that is construction of an artifact that can be tested against operational needs, improved and reused this research will be conducted utilizing a Design Science Research (DSR) approach (Hevner et al., 2004). The pipeline's stages are as follows:

Stage 1 is Data Ingestion and Split: Because of the laptops memory capacity this will be done in small batches. Before we start feature engineering, we will perform the temporal split as previously mentioned followed by an initial investigation of the dataset's class distribution, typology distribution and any problems with the data duplicates, etc.

Stage 2 is Features Engineering: Designing features is the most costly step instead of using characteristics of the underlying transactions the goal here is to engineer some features that represent patterns seen in money laundering as how many transactions an account is making in different time bands, how close the amounts are to reporting threshold amounts which is a proxy for structuring, how many counterparties the account is doing transactions with versus receiving from and how unusual the timing of a transaction is in relative to previous transactions with the account. All of these are calculated using only moving windows of accounts history which is necessary to guarantee that they would remain valid in a production setting.

Stage 3 is Baselines: Get two baselines ready one is a typology based rules based system that uses fan out, velocity and structure rules. It can be adjusted to generate the same number of warnings as the machine learning models for similar operating expenses. The other is a logistic a straightforward type of machine learning with the same capabilities as XGBoost. Another important aspect of this research is the separation of a non ML baseline rules and an ML baseline logistic regression which allows for the separation of the contributions of cost sensitive gradient boosting as XGBoost vs Logistic Regression and learning from data rules vs ML.

Stage 4 is Cost sensitive XGBoost training: A naive XGBoost without class specific modifications a cost sensitive XGBoost using the “*scale_pos_weight = inverse ratio of classes in the dataset*” and an XGBoost trained using oversampled classes using the SMOTE technique (Chawla et al., 2002) will all be built and compared. To maximize the Precision Recall AUC their hyperparameters will be adjusted using grid search and 5 fold cross validation on training data. This will enable us to test our hypothesis regarding the most effective way to address the datasets class imbalance which was identified in the literature review.

Stage 5 is Reason codes and SHAP explanations: TreeSHAP (Lundberg and Lee, 2017) will be utilized to generate explanations for the transactions in the test set using the optimal XGBoost hyperparameters discovered in Stage 3. Any of the 12 reason codes RC will be mapped to the top three attributes of the top K alert. RC 07 structuring transactions just under a reporting threshold will be translated to the same values of transactions just under a threshold in the form of a pattern and RC 01 high transaction velocity in brief window will be the reason code for a significant number of transactions during the last hour. To ensure that only a specific set of reasons codes are applied to alerts of the same type an audit trail will be in place.

Stage 6 is Artefact Production: Essentially, a Jupyter Notebook of the end-to-end processing should include a CSV export version of the alert list as a decision support document for STR along with all the processes required to get from CSV to an alert list, scored and commented on.

3.4. Resources

It’s all open source such as Python and Pandas, NumPy data processing packages will be used along with baselines and K fold Cross Validation from scikit learn, SMOTE from imbalanced learn, classification from XGBoost 2.0, explanation from SHAP 0.45 and visualization from Matplotlib plus Seaborn. The search can be conducted on the preferred device however the pipeline can be executed locally on a 16GB memory PC without a cloud instance.

3.5. How performance will be evaluated

Instead, then focusing only on numbers we aim to assess performance for AML reasons. There are five metrics used the first is the PR AUC (Precision Recall AUC) which is superior to the more well known ROC AUC in situations where the number of transactions is significantly skewed and we are concentrating on the minority class. The recall at a predetermined threshold number of alarms to review the number of actual laundering transactions in the top 0.5%, 1% and 5% of the list of rank ordered alarms comes in second. This is more realistic since in the real world compliance officers can’t just rank and review any alarm that exceeds a given threshold because they have a limited fixed amount of time each week to inspect alarms (Oztas et al., 2024). The spent analyst time per true positive, false positives at the ideal alert count is the third. The fourth is

the consistency of individual reasoning explanations of the percentage of STEM alarms with the same typology that have two of our top three reason codes which represents the consistency of audit explanations (Adadi and Berrada, 2018). The final one is an inference run time meaning it takes four hours to run overnight. Additionally, we shall present the findings by sub typology. It is useful to know that the model is good but weakened by a scatter gather pattern, for example, if someone plans to utilize this tool.

3.6. Ethics

There are a number of ethical considerations regarding privacy as there are no privacy concerns with individual data because the IBM server data is open and synthetic. The Ethics Declaration Form makes note of this as there is a problem with fairness because it is possible to optimize for a minority class in a way that is harmful to a certain kind of consumer, for instance certain types of operations are linked to laundering hotspots. The dataset cannot be analyzed for racial, gender and other disparity because it lacks any demographic relationships. The research will emphasize this and various transaction types will be highlighted by the slice analysis at the typology level.

To prevent using XplainTM as a black box deployment must be done with extreme caution. A human should be able to understand an alert from it. This is consistent with both Article 22 on automated decisions of the General Data Protection Regulation (GDPR) and the suggested human in the loop method of Adadi and Berrada (2018). The reason codes goal is to help people in order to ensure reproducibility as all metrics will be published on the held out test data because our code will be made public in the artifact and the temporal split will prevent data leaks.

3.7. Thesis Schedule

The time periods from May, 18 2026 to August 7, 2026 are displayed in the Gantt chart below:

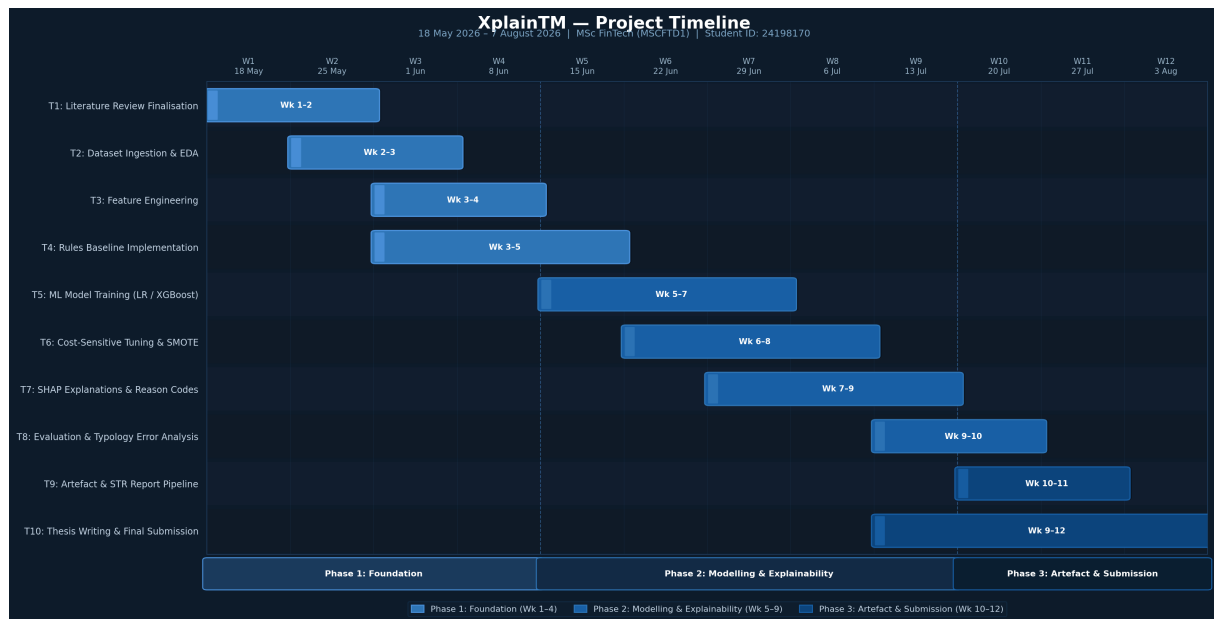


Figure 1. XplainTM Project Gantt Chart (18 May 2026 – 7 August 2026)

1.5-week schedule is the appropriate tasks in the appropriate sequence and concurrently the plan is divided into three stages:

Weeks 1-4 (May 18–June 14) represent Phase 1 as this week is dedicated to preparation. The final two weeks of reading and developing the theory are now underway. The time split for the train, validate and test is applied once the data has been imported and cleaned. In order to have feature engineering data pipeline and baseline software rule engine by week 4 and be prepared to create models the third week is dedicated to feature engineering and building the baseline software.

Phase 2 runs from week 5 to week 9 (June 15–July 19) as this phase involves a lot of codes in artifact. We will train the XGBoost models and complete baseline tuning in week 5. In weeks 6 through 8 we will compare the cost sensitive and SMOTE models and in week 7 we will start experimenting with our SHAP explanations. This makes it possible for a model to integrate the SHAP explanations and the model selection takes into account the consensus of the reason codes examined. Both the evaluation study and the error analysis based on typology subgroups are completed in week 9.

Weeks 10 through 12 (July 20–August 7) comprise Phase 3 at that period we concentrate on the Jupyter Notebook, CSV file and artifact setup in weeks 10 and 11. Report thesis will begin in week 9 and be completed by August 7th. Proofreading and submission take place in week 12.

References

- Adadi, A. and Berrada, M. (2018). Peeking inside the black-box: a survey on explainable artificial intelligence (XAI). *IEEE Access*, 6:52138–52160. Scopus Q1; CiteScore 9.4.
- Alarab, I., Prakoonwit, S., and Nacer, M. I. (2022). Competence of graph convolutional networks for anti-money laundering in Bitcoin blockchain. In *Proceedings of the 2022 5th International Conference on Machine Learning and Machine Intelligence (MLMI 2022)*, pages 1–6. Google Scholar citations: 290+.
- Altman, E., Blumenstock, J., Donahoe, E., Fowers, M., Germann, D., Hamilton, E., Helm, I., Ismail, M., Kaul, V., Laptev, N., et al. (2023). Realistic synthetic financial transactions for anti-money laundering models. In *Advances in Neural Information Processing Systems (NeurIPS 2023) — Datasets and Benchmarks Track*. CORE A*; Google Scholar citations: 180+.
- Arner, D. W., Barberis, J., and Buckley, R. P. (2017). FinTech, RegTech and the reconceptualization of financial regulation. *Northwestern Journal of International Law and Business*, 37(3):371–413. Google Scholar citations: 1,800+.
- Bahnsen, A. C., Aouada, D., and Ottersten, B. (2016). Example-dependent cost sensitive decision trees. *Expert Systems with Applications*, 42(19):6609–6619. Scopus Q1; Impact Factor 8.7.
- Chawla, N. V., Bowyer, K. W., Hall, L. O., and Kegelmeyer, W. P. (2002). SMOTE: synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357. Scopus Q1; Google Scholar citations: 21,000+.
- Chen, T. and Guestrin, C. (2016). XGBoost: a scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794. CORE A*; Google Scholar citations: 45,000+.
- Financial Action Task Force (2022). Money laundering. <https://www.fatf-gafi.org/en/topics/money-laundering.html>.
- Hevner, A. R., March, S. T., Park, J., and Ram, S. (2004). Design science in information systems research. *MIS Quarterly*, 28(1):75–105. Scopus Q1; Google Scholar citations: 11,500+.
- Jensen, P. A. and Iosifidis, A. (2023). Anti-money laundering alert optimization using machine learning with graphs. arXiv preprint arXiv:2112.07508. Proceedings of the AAAI Workshop on AI for Financial Services. CORE A (AAAI); Google Scholar citations: 60+.
- LexisNexis Risk Solutions (2022). True cost of financial crime compliance study — Global Report. <https://risk.lexisnexis.com/insights-resources/research/true-cost-of-financial-crime-compliance>.

- Lundberg, S. M. and Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems (NeurIPS 2017)*, volume 30, pages 4765–4774. CORE A*; Google Scholar citations: 15,000+.
- Oztas, B., Cetinkaya, D., Adedoyin, F., Budka, M., Aksu, G., and Dogan, H. (2024). Transaction monitoring in anti-money laundering: a qualitative analysis and points of view from industry. *Future Generation Computer Systems*, 159:161–171. Scopus Q1; Impact Factor 8.95.